

3 (a) Phase difference = $\frac{2\pi t}{T} = \frac{2\pi(2.5-1.5)}{3} = \frac{2\pi}{3} = 120^\circ$ **A1**

(b) At $t = 0.75$ ms, resultant displacement = $s_Y + s_Z = 4.0 - 1.0 = 3.0 \mu\text{m}$ **A1**

(c) $I \propto A^2$ **C1**

Thus $I = k(4)^2$

and $I_Z = k(2)^2 = 0.25I$ **A1**

(d) $v = f\lambda = \frac{\lambda}{T}$ **C1**

so $\lambda = vT = 330(3.0 \times 10^{-3}) = 0.99 \text{ m}$ **A1**

4 (a) (i) Current in R_1 or $R_2 = 0.30 + 0.30 = 0.60 \text{ A}$ **C1**